

DES PUNE UNIVERSITY
School of Engineering and Technology
Computer Engineering and Technology
Program: B.Tech. Computer Science and Engineering

Academic Year: 2025-26	Year: Second Year	Term: I
Roll No.: 1012411011	Name: Riddhee Kulkarni	
Subject: Java		
Assignment No.: 9	Title: Hospital Management System	
Date: 30/09/2024		

Aim: Write a program to connect to a database using JDBC, execute a simple SQL query, and display the results. Implement JDBC connectivity for Hospital management system. (No need of GUI creation for this system)

Theory:

JDBC Connectivity:

JDBC (Java Database Connectivity) is an API in Java that allows applications to connect to and interact with relational databases. It provides a standard interface for sending SQL queries, updating data, and retrieving results. JDBC uses drivers specific to each database (like MySQL, Oracle, PostgreSQL) to establish the connection.

SQL Query:

SQL (Structured Query Language) is used to interact with databases. Through queries, you can:

Retrieve data (using SELECT)

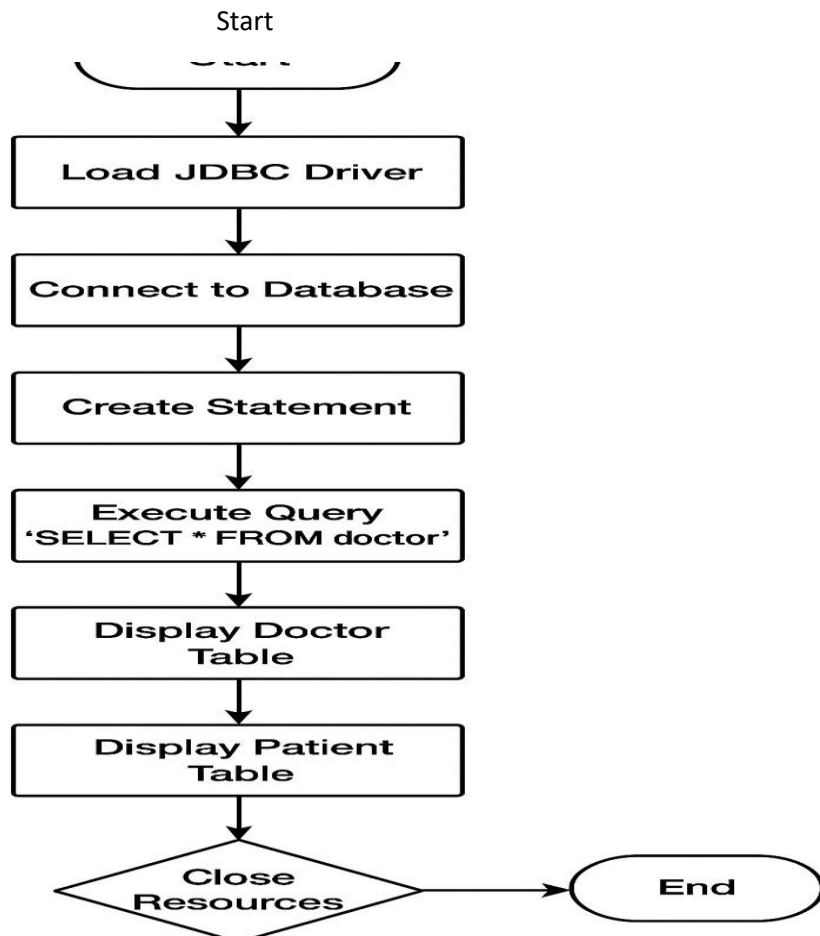
Insert new records (INSERT)

Update existing records (UPDATE)

Delete records (DELETE)

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Flowchart:



Code:

```
package Assignment9;

import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.ResultSet;
import java.sql.Statement;

public class HospitalManagementSystem
{
    public static void main(String[] args)
    {
        Connection conn = null;
        Statement stmt = null;
        ResultSet rs = null;

        try
        {
            // Load the Oracle JDBC driver
            Class.forName("oracle.jdbc.driver.OracleDriver");

            // Connect to the Oracle database using service name
            // Replace ORCLPDB1 with your actual service name from lsnrctl status
            String url = "jdbc:oracle:thin:@localhost:1521:xe";
            String user = "system"; // if connecting as SYSDBA
            String password = "despu";

            conn = DriverManager.getConnection(url, user, password);
```

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```
// Create a statement object
stmt = conn.createStatement();

// Fetch doctor data
rs = stmt.executeQuery("SELECT * FROM doctor");

System.out.println("Doctor Table:");
System.out.println("ID | Name | Specialization");
while (rs.next())
{
    int docId = rs.getInt("doc_id");
    String docName = rs.getString("doc_name");
    String specialization = rs.getString("specialization");

    System.out.println(docId + " | " + docName + " | " + specialization);
}

// Close previous ResultSet before running another query
rs.close();

// Fetch patient data
rs = stmt.executeQuery("SELECT * FROM patient");

System.out.println("\nPatient Table:");
System.out.println("ID | Name | Age | Disease");
while (rs.next())
{
    int patId = rs.getInt("pat_id");
    String patName = rs.getString("pat_name");
    int patAge = rs.getInt("pat_age");
    String disease = rs.getString("disease");

    System.out.println(patId + " | " + patName + " | " + patAge + " | " + disease);
}
}
catch (Exception e)
{
    System.out.println("Error: " + e);
}
finally
{
    // Close resources in reverse order
    try
    {
        if (rs != null)
        {
            rs.close();
        }
    }
    catch (Exception e)
    {
    }
}

try
```

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```
{
    {
        {
            if (stmt != null)
            {
                stmt.close();
            }
        }
        catch (Exception e)
        {
        }
    }

    try
    {
        if (conn != null)
        {
            conn.close();
        }
    }
    catch (Exception e)
    {
    }
}
}
```

DOCTOR

Create

Table

Data

Indexes

Model

Constraints

Grants

Statistics

UI Defaults

Triggers

Dependencies

SQL

Add Column

Modify Column

Rename Column

Drop Column

Rename

Copy

Drop

Truncate

Create Lookup Table

Column Name	Data Type	Nullable	Default	Primary Key
DOC_ID	NUMBER(5,0)	No	-	1
DOC_NAME	VARCHAR2(50)	No	-	-
SPECIALIZATION	VARCHAR2(50)	Yes	-	-

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PATIENT

Create

TableDataIndexesModelConstraintsGrantsStatisticsUI DefaultsTriggersDependenciesSQL

Add ColumnModify ColumnRename ColumnDrop ColumnRenameCopyDropTruncateCreate Lookup Table

Column Name	Data Type	Nullable	Default	Primary Key
PAT_ID	NUMBER(5,0)	No	-	1
PAT_NAME	VARCHAR2(50)	No	-	-
PAT_AGE	NUMBER(3,0)	Yes	-	-
DISEASE	VARCHAR2(50)	Yes	-	-

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Output:

Doctor Table:

ID	Name	Specialization
102	Dr. Mehta	Neurology

Patient Table:

ID	Name	Age	Disease
201	Riddhee Kulkarni	25	Flu

Conclusion:

The program connects to an Oracle database, retrieves doctor and patient data, displays it, and closes all resources properly.